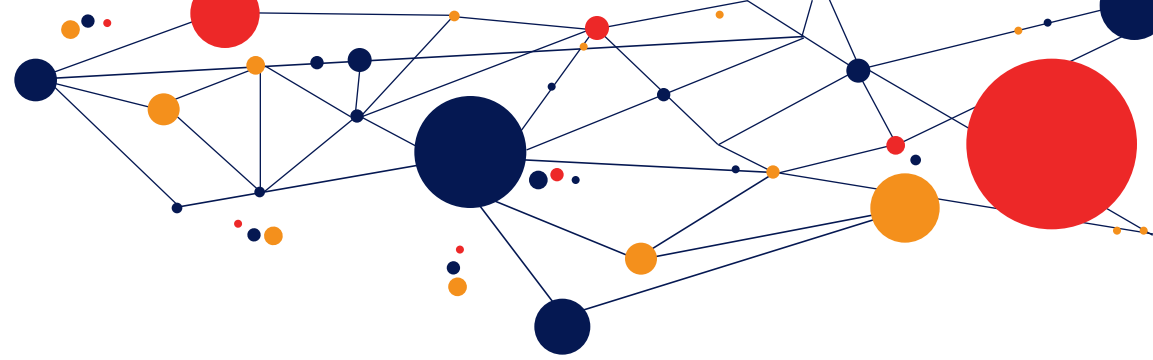


BSE INDEX SERVICES PVT. LTD.

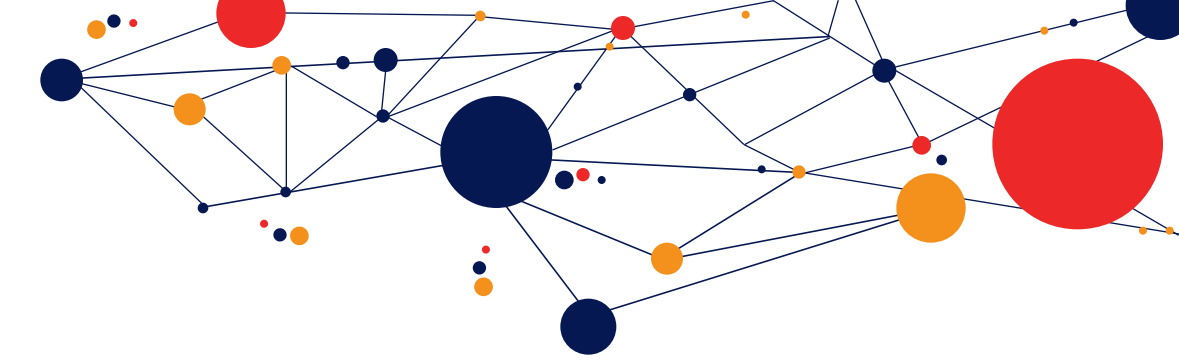
***Automation & Analytical Enhancement of Free Float Factor Identification
& Creation of BSE Sector Rotation Index (BSRI)***

Project:

Product Manager and Data Analyst Intern

Need:





Earlier Methodology at BSE to Identify FreeFloat Shares

Refer previous quarter FFF values; select stocks to check, if difference is $> 2\%$

Visit BSE Website for each selected stock (800+ per quarter, 3 min / stock)

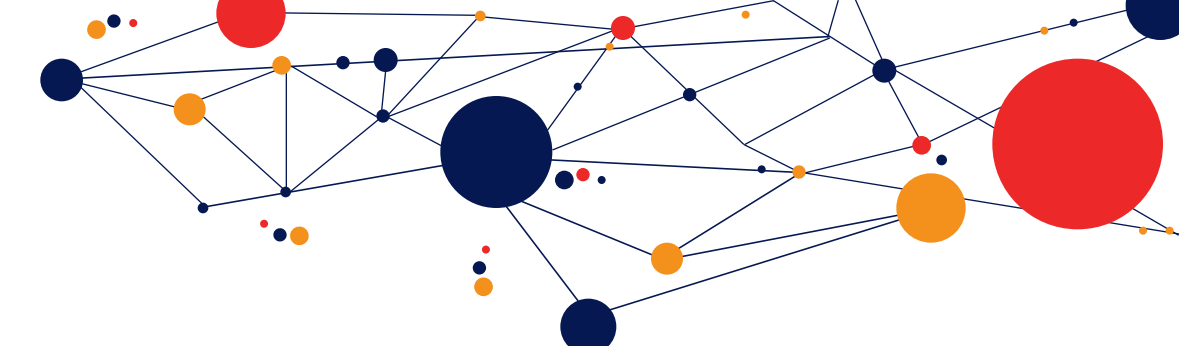
Visit annual reports for exceptional cases for analysis (50 cases, 10 min/stock)

Mark each NEW shareholder as Strategic, Promotor OR Public Shareholders

Calculate new free float values by inputting the changes and running formulas

Free Float Shares?

Root Cause of Partial Structured Shareholding Pattern : *Many Companies try to maximize their Free Float Shares by declaring Strategic Promoters Shareholders as Public*



Objectives (Free Float Factor Model)

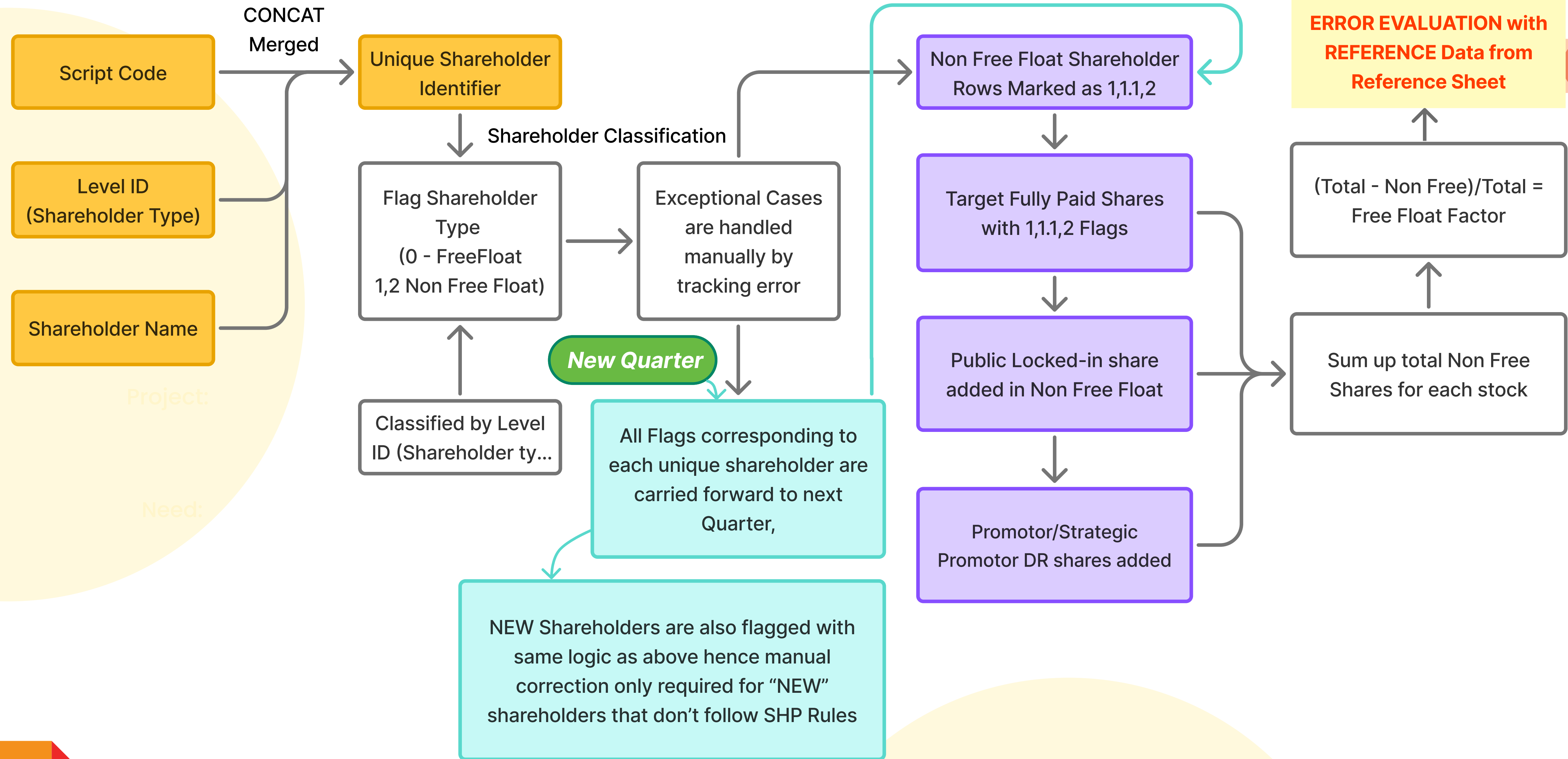
- *Standardize Free Float Factor calculations across BSE indices using automation*
- *Reduce operational time and manual errors in index management and reporting*

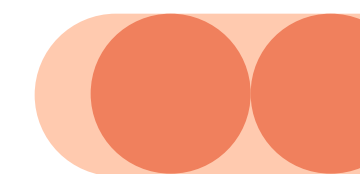
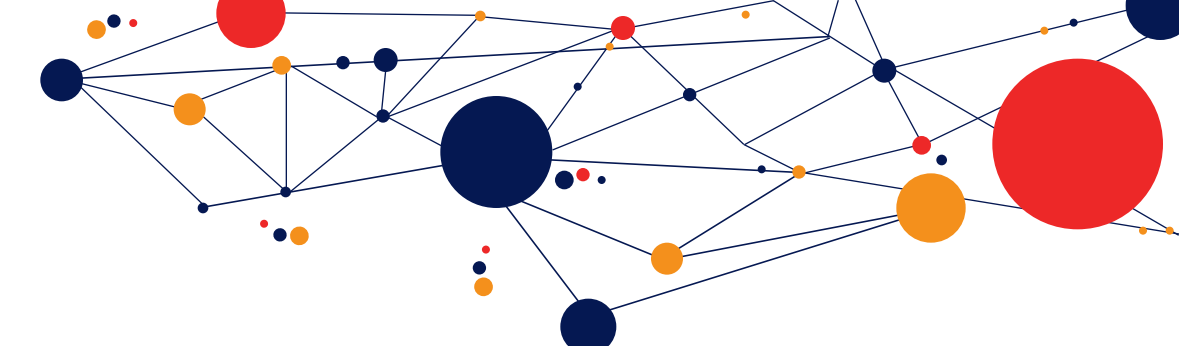
Methodology (Free Float Factor Model)

- *Excel-SQL model to automate shareholder classification and Free Float Factor (FFF) computation*
- *Rule-based flags for Promoter, Strategic, Doubtful, and Public shareholders*
- *Quarterly data validation against historical records to ensure compliance and accuracy*

Need:

Methodology (Free Float Factor Model)



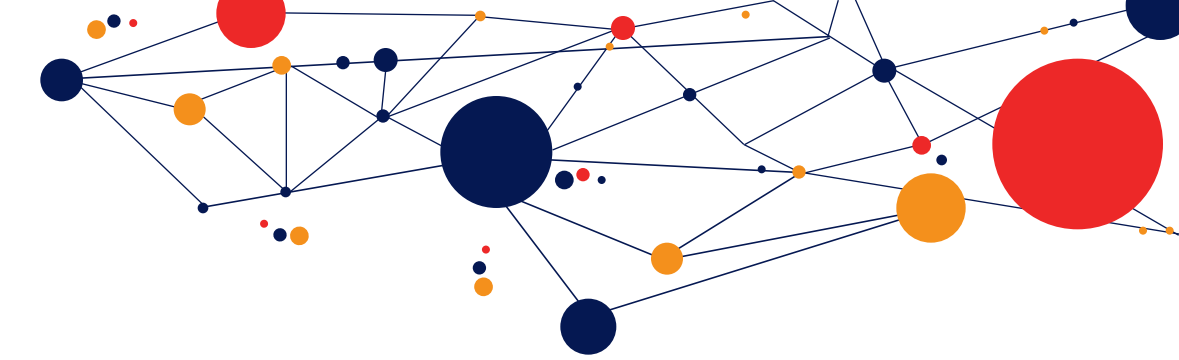


Results & Impact (FFF Model)

- ***Boosted daily BSE trade volume by ₹700 Cr (0.7%)***
- ***Reduced process time: 20 days/quarter x 3 FTEs → 1 day/quarter x 1 FTE (80x faster)***
- ***Eliminated 1–5% error in Free Float shares for 300 stocks out of 4,800+***
- ***Increased Free Float Shares Weightage in every BSE Index (Even Sensex)***
- ***Enabled accurate root-cause tracing and compliance with global standards***

Need:

PROJECT 2 : BSE SECTOR ROTATION INDEX



Pull & download all Index data (Date, Values, Constituents, etc) internally

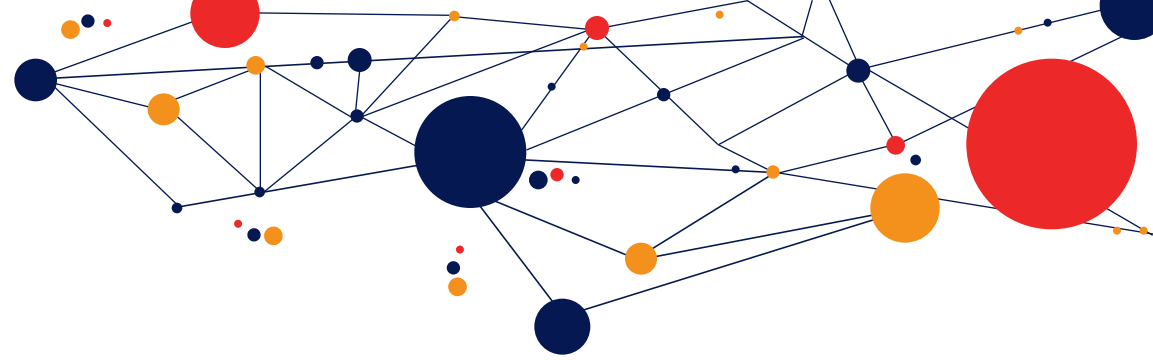
Pull & download all Index stocks constituent data (Sector, Date, Price, Weight)

Copy Paste both data in a Standard Excel formula template

Calculate Index Performances, Index details for each month for 101 Index

Create Sales PPT on MS Powerpoint before meeting Wealth Management Firms

Organize all 101 indices every month, and clear them next month



Objectives (Sales Deck Automation)

- *Develop automated generation of branded, analytics-rich client presentations*
- *Reduce manual reporting workload for Sales team*
- *Enhance client delivery speed and consistency across 101 indices*

Methodology (Sales Deck Automation)

- *Decide what data needs to be shown to client*
- *Build presentation design using Figma*
- *Collaborate with Front-End Team to build the Presentation ready for PDF conversion*
- *Collaborate with Data team to arrange required raw data*
- *Collaborate with Python & SQL team utilizing raw data for final calculations*
- *Collaborate with Backend team to bind final data with frontend and make the PDFs downloadable on a secure web portal just by selecting Index type and Month*

BSE MIDCAP

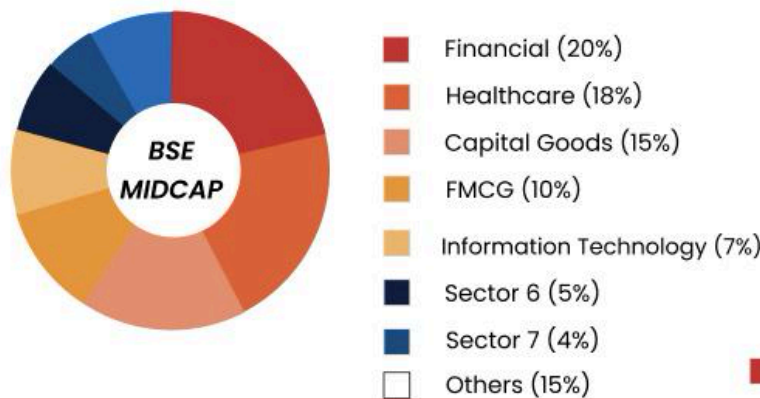
Overview

The BSE SENSEX is India's most tracked bellwether index. It is designed to measure the performance of the 30 largest, most liquid and financially sound companies across key sectors of the Indian economy that are listed at BSE Ltd. It was launched on 1st January 1986.

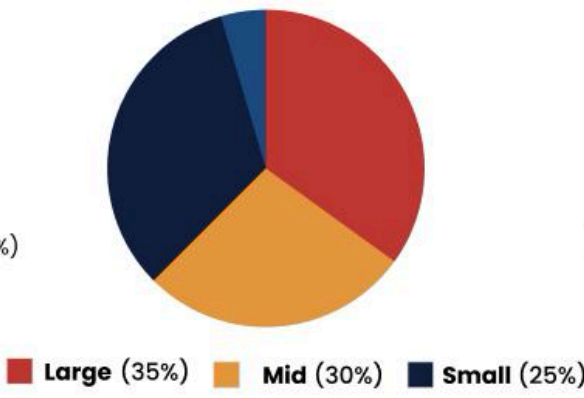
Quick Facts

Launch Date	1st January 1986
First Value Date	3rd April 1979
Base Value	100
Number of Constituents	30
Reconstitution Frequency	Semiannually in June and December
Weighting Method	Float adjusted market cap weighted
Index Universe	BSE 100
Calculation Currencies	INR, USD

Sectorial Composition of BSE MIDCAP



Market Cap Size Coverage



Fundamentals

PE	20
PB	24
Dividend Yield %	14

Top Constituents by Weightage



BSE MIDCAP

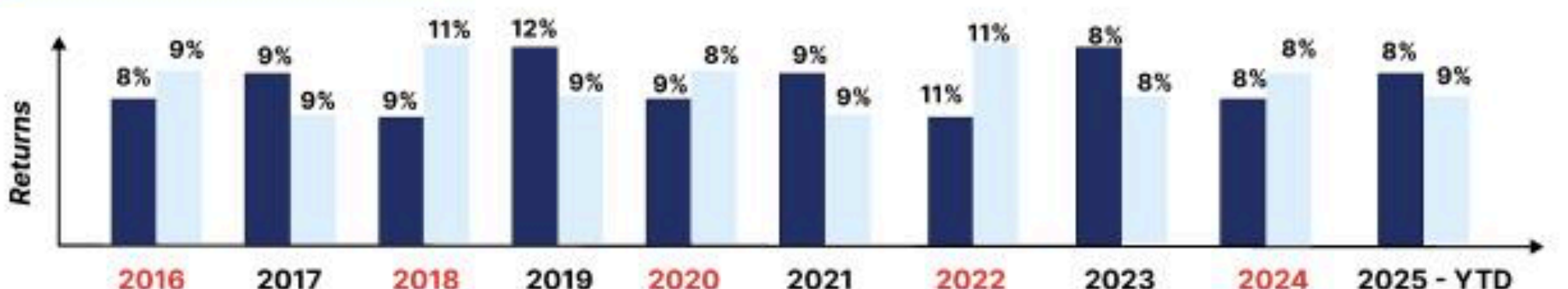
Returns

Index Level	Absolute Returns			Annualized			
	1 Month	3 Month	YTD*	1 Year	3 Years	5 Years	10 Years
BSE Midcap							
Sensex							

Risk Returns

Index Level	Annualized Risk			Annualized Risk - Adjusted Returns			
	1 Month	3 Month	YTD*	1 Year	3 Years	5 Years	10 Years
BSE Midcap							
Sensex							

Comparision of BSE MIDCAP Calender Year Returns with BSE SENSEX



BSE MIDCAP

Rolling Returns

	1 Year	3 Years	5 Years	7 Years	10 Years
Average					
Median					
Minimum					
Maximum					
Total No.of Observation (Days)					
Positive Returns Observation (Days)					
Negative Returns Observation (Days)					

BSE MIDCAP

Performance

Performance during Market Cycles

From	To	Event	Returns (Absolute)
11-Sep-05	31-Dec-07	Bull	143.35%
11-Sep-05	11-Sep-05	Recovery	143.35%
11-Sep-05	11-Sep-05	Bear	143.35%
11-Sep-05	11-Sep-05	Bull	143.35%
11-Sep-05	11-Sep-05	Bull	143.35%
11-Sep-05	11-Sep-05	Recovery	143.35%
11-Sep-05	11-Sep-05	Bull	143.35%
11-Sep-05	11-Sep-05	Bull	143.35%
11-Sep-05	11-Sep-05	Recovery	143.35%
11-Sep-05	11-Sep-05	Bull	143.35%
11-Sep-05	11-Sep-05	Recovery	143.35%
11-Sep-05	11-Sep-05	Bear	143.35%

Performance during Global Crisis

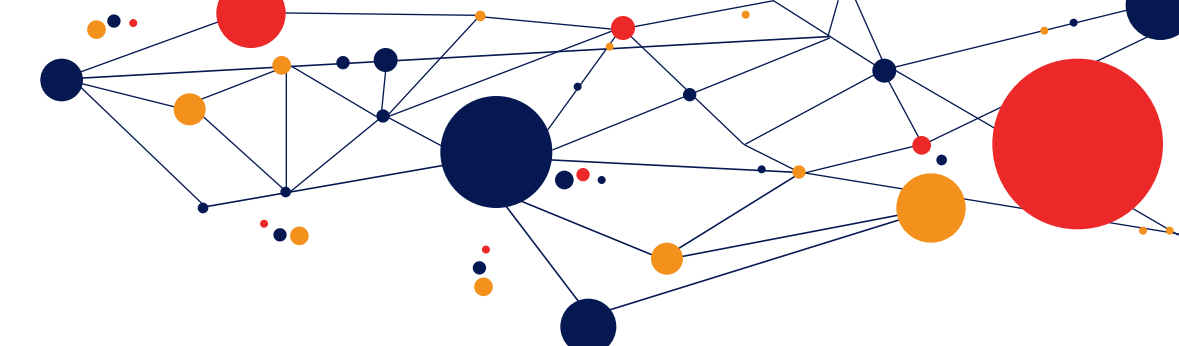
Event	From	To	Returns during Crisis (Absolute)	Returns after 12 Month of Crisis	Returns after 12 Month + During Crisis
Global Financial Crisis	11-Sep-05	31-Dec-07	143.35%	143.35%	143.35%
Taper Tantrum	11-Sep-05	31-Dec-07	143.35%	143.35%	143.35%
Yuan Devaluation	11-Sep-05	31-Dec-07	143.35%	143.35%	143.35%
Covid 19 Pandemic	11-Sep-05	31-Dec-07	143.35%	143.35%	143.35%

Covid-19 performance (Recent Crisis)

Event	From	To	Returns during Crisis (Absolute)
Year till Covid Crash	11-Sep-05	31-Dec-07	143.35%
Covid Crash	11-Sep-05	31-Dec-07	143.35%
Bottom to Last Month End	11-Sep-05	31-Dec-07	143.35%
Since Covid to Last Month End	11-Sep-05	31-Dec-07	143.35%

Drawdown Return Comparision

BSE MIDCAP	
Draw Down Date	Maximum Drawdown
BSE SENSEX	
Draw Down Date	Maximum Drawdown

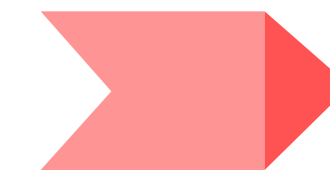


Results (Sales Deck Automation)

- *90% reduction in manual prep; saved 30+ hrs/month for Sales team*
- *Client turnaround time improved by 20%*
- *Consistent, error-free, branded presentations now standard at BSE*
- *Portal adopted as sole tool for sales reporting across all indices*

Project:

Need:



THANK YOU

Project:

Need:

